

Memory Under Pressure

How Fixed-Size State Remembers, Updates, Interferes and Forgets Across Long Sequences

Competition Track: Pathway Track (Explain the Frontier)

Project: Memory Under Pressure

Interactive Artifact: <https://memory-under-pressure-flax.vercel.app>

Source Repository: github.com/lshantTripathi/memory-under-pressure

1. Why Long Context Creates a Memory Problem

In contemporary Large Language Models (LLMs) built on standard multi-head attention (the Transformer architecture), context retention relies on an external buffer known as the **Key-Value (KV) Cache**. During autoregressive token generation, every newly processed token generates a key vector $k_t \in \mathbb{R}^d$ and a value vector $v_t \in \mathbb{R}^d$ for each attention layer and head. Because standard attention computes pairwise Softmax dot products between the current query and all historical keys, none of these past vectors can be discarded.

As a consequence, the physical memory required to maintain this history grows strictly linearly with sequence length T :

$$\text{KV-Cache Footprint} = 2 \times T \times d \times L \times \text{bytes_per_element}$$

where T is sequence length, d is head dimension, and L is layer count. For modern context windows spanning 32,000 to 1,000,000+ tokens, this linear growth in cached-token memory presents severe physical bottlenecks:

- **GPU High-Bandwidth Memory (VRAM) Exhaustion:** Maintaining gigabytes of cached activations per concurrent user session exhausts available accelerator memory, bounding serving concurrency.
- **Memory Bandwidth Bottleneck:** In autoregressive decoding, loading hundreds of megabytes of KV vectors from HBM into on-chip SRAM for every single generated token bounds inference throughput to memory bandwidth rather than compute FLOPs.
- **Reasoning Token Overhead:** When frontier models solve complex tasks via extended Chain-of-Thought (CoT), generating thousands of intermediate reasoning tokens compounds KV cache accumulation and drastically elevates serving costs.

This tension motivates a fundamental architectural question: *Can a neural system process arbitrarily long token sequences using a memory representation that remains fixed with respect to sequence length T for a fixed model configuration?*

2. The Central Falsifiable Claim

The educational premise of this project is organized around a single falsifiable technical hypothesis:

"A fixed-size evolving state can process sequences whose duration grows without allocating a new memory slot for every token, but limited state capacity can cause interference and forgetting."

This statement contains two reciprocal assertions:

- **The Promise:** A recurrent model can maintain an internal working memory representation whose size is bounded independently of sequence length T (e.g., an explicit $d \times d$ matrix state requiring $O(d^2)$ entries), avoiding per-token buffer allocation.
- **The Physical Constraint:** Because the representation size does not grow with T , incoming facts must share the same finite-dimensional subspace. This superposition inevitably produces cross-talk interference and forces information trade-offs as sequence length grows relative to the available state.

3. A Minimal Associative Memory Model

To make the algebraic mechanics of evolving states inspectable in real time, the project implements an **intentionally small educational associative-memory model inspired by the broader family of recurrent and state-based architectures** (such as Linear Transformers, RetNet, RWKV, and Fast Weight Programmers).

The model maintains an explicit 2D matrix state $S_t \in \mathbb{R}^{d \times d}$. At each discrete timestep $t \in \{1, 2, \dots, T\}$, the system receives a token representation decomposed into an association pair:

- **Key vector:** $k_t \in \mathbb{R}^d$, normalized such that $\|k_t\| = 1$, indicating the address or attribute slot.
- **Value vector:** $v_t \in \mathbb{R}^d$, normalized such that $\|v_t\| = 1$, encoding the stored content.

The state matrix updates according to a generalized outer-product recurrent rule:

$$S_t = \lambda S_{t-1} + \eta (v_t \otimes k_t^T)$$

where S_t is the recurrent state matrix initialized to $S_0 = 0$, $\lambda \in [0, 1]$ is a recency decay factor, $\eta > 0$ is the update rate (default $\eta = 1.0$), and $v_t \otimes k_t^T = v_t k_t^T$ denotes the outer product of value and key vectors, producing a rank-1 matrix of size $d \times d$ whose (i, j) entry is $v_{t,i} k_{t,j}$.

Expanding the recurrence from $t = 1$ to t yields the closed-form representation:

$$S_t = \sum_{\tau=1}^t \lambda^{t-\tau} \eta (v_\tau k_\tau^T)$$

4. What the State Matrix Actually Stores

Standard Transformer models store a list of isolated token vectors: $[k_1, \dots, k_t]$ and $[v_1, \dots, v_t]$. In contrast, an associative recurrent state stores the **linear superposition of outer products**.

Each individual cell $S_{i,j}$ in the matrix tracks the accumulated correlation between the i -th feature of past values and the j -th feature of past keys:

$$S_{i,j}^{(t)} = \sum_{\tau=1}^t \lambda^{t-\tau} \eta \cdot v_{\tau,i} \cdot k_{\tau,j}$$

Because matrix addition is commutative and associative, the state acts as an integrated memory canvas:

- Information is distributed across the entire matrix rather than sequestered in distinct token slots.
- The Frobenius norm $\|S_t\|_F = (\sum_{i,j} S_{i,j}^2)^{1/2}$ measures accumulated memory energy.
- Without decay ($\lambda = 1.0$), the norm grows monotonically as new facts arrive. With decay ($\lambda < 1.0$), the norm stabilizes toward a steady-state bound $\|S_\infty\|_F \leq \eta / (1 - \lambda)$.

5. Retrieval: Asking the State for a Fact

To retrieve the value associated with a specific query key probe $q \in \mathcal{H}^d$ (with $\|q\| = 1$), the model computes a standard matrix-vector product:

$$\hat{v} = S_t q$$

Substituting the closed-form expansion of S_t

$$\hat{v} = (\sum_{\tau=1}^t \lambda^{t-\tau} \eta v_\tau k_\tau^T) q = \sum_{\tau=1}^t \lambda^{t-\tau} \eta v_\tau (k_\tau^T q)$$

Notice the associative re-grouping: because matrix multiplication is associative, $(v_\tau k_\tau^T) q = v_\tau (k_\tau^T q)$. The scalar dot product $k_\tau^T q$ measures cosine similarity between the query probe q and each historical key k_τ . The retrieved vector $\hat{v} \in \mathcal{H}^d$ is thus a linear combination of all stored value vectors, weighted by key similarity and recency attenuation. Discrete fact classification is performed by nearest-neighbor codebook lookup:

$$\text{predicted_value} = \arg \max_{c \in V} (\hat{v}^T v_c / (\|\hat{v}\| \|v_c\|))$$

6. Signal, Noise and Interference

When querying memory for a specific target fact injected at step t_{target} with key k_{target} and value v_{target} the output decomposes analytically into two orthogonal parts:

$$\begin{aligned} \hat{v} &= \lambda^{t-t_{\text{target}}} \eta (k_{\text{target}}^T q) v_{\text{target}} + \sum_{\tau \neq t_{\text{target}}} \lambda^{t-\tau} \eta (k_\tau^T q) v_\tau \\ \hat{v} &= [\text{Target Signal Vector } s] + [\text{Cross-Talk Noise Vector } n] \end{aligned}$$

- **Target Signal (s):** The component aligned with v_{target} . For an exact probe ($q = k_{\text{target}}$), signal norm is $\|s\| = \lambda^{t-t_{\text{target}}} \eta$.
- **Cross-Talk Noise (n):** The vector sum of all other historical values, each scaled by $k_\tau^T q$. Whenever historical keys are non-orthogonal to the query, a fraction of their value leaks into the retrieval.

We quantify retrieval fidelity via the Signal-to-Noise Ratio (SNR) in decibels:

$$\text{SNR}_{\text{dB}} = 10 \cdot \log_{10} (\|s\|^2 / (\|n\|^2 + \epsilon))$$

- **$SNR > +15\text{ dB}$ (High Fidelity):** Target signal completely dominates noise. Nearest-neighbor decoding reliably identifies the correct fact.
- **$0\text{ dB} \leq SNR \leq +10\text{ dB}$ (Degraded / Borderline):** Noise distorts the retrieved vector. Classification margins shrink, risking misclassification.
- **$SNR < 0\text{ dB}$ (Interference Collapse):** Cross-talk noise exceeds signal ($\|n\| > \|s\|$). The model predicts a spurious distractor fact.

7. Why Capacity Depends on State Dimension

Linear algebra's dimension theorem governs the capacity of fixed-size associative states:

- **Orthogonal Subspace Bound:** In $\mathcal{H}^d$, at most d non-zero vectors can be mutually orthogonal ($k_{-i}^T k_{-j} = 0$ for all $i \neq j$).
- **Zero Cross-Talk Regime ($T \leq d$):** If $T \leq d$ and keys form an orthonormal set, then for $q = k_{target}^T k_{-\tau}^T$, $q = 0$ for all $\tau \neq target$. The noise vector is identically zero: $n = 0 \Rightarrow SNR \rightarrow \infty$.
- **Superposition Saturation ($T > d$):** When sequence length T exceeds dimension d , incoming keys are guaranteed to be linearly dependent on previous keys. Cross-talk noise becomes mathematically unavoidable.
- **Rank Bottleneck:** The matrix $S_{-t} \in \mathcal{H}^{d \times d}$ has rank at most $\min(d, T)$, saturating at d . Further updates overwrite existing directional subspaces.

8. Decay and Forgetting

In this toy recurrence, setting $\lambda < 1.0$ attenuates older state contributions and helps prevent unbounded accumulation. However, it imposes an exponential recency amnesia. A fact stored at step τ has its signal multiplied by λ at every subsequent step:

$$\|s(\tau, t)\| = \lambda^{t-\tau} \eta$$

For example, with $\lambda = 0.90$, an item stored 30 steps ago retains only $0.90^{30} \approx 0.042$ (a 95.8% signal reduction). While recent items remain crisp, early context is effectively erased.

9. Key Correlation and Information Leakage

In real-world text, token embeddings are clustered on semantic submanifolds with non-zero mutual cosine similarities ($\rho = k_{-i}^T k_{-j} > 0$). When keys correlate, distractor tokens leak a fraction $\rho_{-\tau}$ of their values directly into the output. Correlated keys can induce retrieval leakage even when the state dimension is large, depending on the update and retrieval conditions. Usable memory capacity is thus constrained not only by dimension d , but also by the geometric angular separation of input keys.

10. Interactive Experiments in Memory Under Pressure

The repository's interactive simulator exposes 6 deterministic presets (using the Mulberry32 PRNG with seed 42) that demonstrate these theoretical phase transitions:

Preset Name	Configuration	Theoretical Prediction	Empirical Outcome
Clean Baseline	$T=5, d=16, \lambda=1.00, \rho=0.00$	Orthogonal regime ($T < d$). Keys mutually perpendicular.	Noise = 0.000, SNR = +90.00 dB (ceiling), 100% exact retrieval.
Capacity Threshold	$T=16, d=16, \lambda=1.00, \rho=0.10$	Dimension bound ($T = d$). Matrix approaches full rank.	Noise = 0.452, SNR = +6.90 dB, target retrieved with reduced margin.
Severe Saturation	$T=60, d=8, \lambda=1.00, \sigma=0.05$	Superposition saturation ($T = 7.5 d$). Noise exceeds signal.	Noise = 2.648, SNR = -8.46 dB, retrieval fails (predicts distractor).
Selective Decay	$T=35, d=16, \lambda=0.92, \rho=0.00$	Early fact ($t=1$) attenuated by $0.92^{34} \approx 0.059$.	Signal attenuated by 94.1%, SNR = -16.27 dB, recency amnesia.
Key Collision	$T=20, d=16, \lambda=1.00, \rho=0.70$	Correlated keys leak energy regardless of state size.	Noise = 3.142, SNR = -9.94 dB, severe directional interference.
Break-It Stress	$T=250, d=8, \lambda=0.99, \sigma=0.15$	Catastrophic overload: 250 facts in 8×8 matrix.	Empirical phase-transition curve shows accuracy drop from 100% to <10%.

11. The Failure Cases

The project explicitly documents the architectural limitations of linear associative memory:

- Finite-Rank Saturation:** An associative state matrix $S \in \mathcal{R}^{d \times d}$ has rank at most d . A finite matrix state cannot represent arbitrarily many independent associations without interference as the number of stored associations grows relative to the available state.
- The Leaky State Dilemma:** In this recurrence, setting $\lambda < 1.0$ attenuates older state contributions and helps prevent unbounded state accumulation over long sequences. However, this creates an exponential forgetting horizon where signals scale as λ^t , inducing recency amnesia.
- Quasi-Orthogonality in Real Text:** Natural language embeddings occupy dense semantic clusters, causing cross-talk far earlier than idealized orthogonal synthetic benchmarks.
- Linear vs. Non-Linear Selectivity:** Unlike Softmax attention which exponentiates dot products to suppress distractors, linear associative recall sums projections directly, allowing small noise components to accumulate.
- Educational Toy vs. Frontier Architectures:** The simulator uses an explicit 2D matrix of size $d \times d$ to make linear algebra inspectable. Production architectures employ multi-layer deep states, non-linear activation functions, input-dependent gating, and multi-head projections.

12. Connection to Dragon Hatchling (BDH)

How does frontier AI research address the representational limits of linear recurrence? In 2025, researchers at Pathway introduced Dragon Hatchling (BDH, arXiv:2509.26507). BDH is a bio-inspired Post-Transformer

architecture designed to bridge the gap between artificial attention mechanisms and neuroscience models of the brain.

It is vital to understand the pedagogical relationship:

- **The toy model in this repository is NOT official BDH.** The repository uses a minimal associative outer-product matrix to teach foundational concepts of fixed-state memory, subspace capacity, and interference.
- **BDH is a much richer architecture:** Instead of an unconstrained dense outer-product matrix, BDH models the network as a scale-free graph of locally interacting “neuron particles.”
- **Hebbian Plasticity ($Q = K$):** In biological neuroscience, Donald Hebb’s rule states: “*Neurons that fire together, wire together.*” BDH constrains queries and keys to identical projections ($Q = K$) derived from sparse, non-negative activations (ReLU). The resulting attention matrix is directly interpretable as dynamic synaptic weights.
- **In-Context Plasticity:** During inference, BDH updates synaptic connections dynamically without backpropagation or weight fine-tuning, allowing the network to adapt to demonstration streams while maintaining a fixed internal state representation.

13. BDH-CQ and Demonstration-Based Reasoning

In August 2026, Pathway researchers introduced BDH-CQ (arXiv:2608.09888), extending Dragon Hatchling to long-horizon demonstration learning and iterative reasoning.

In conventional LLMs, complex reasoning requires generating extended verbal Chain-of-Thought (CoT) text tokens. Each generated token must be appended to the KV cache, driving inference cost and latency up quadratically with reasoning depth. BDH-CQ introduces two distinct innovations:

1. **Recurrent In-Context Demo Ingestion:** In-context demonstrations update the model’s recurrent synaptic memory at inference time without gradient fine-tuning.
2. **Latent Workspace Recurrence:** Rather than emitting hundreds of verbal tokens to solve multi-step reasoning queries, BDH-CQ performs iterative recurrence directly within high-dimensional latent space ($z_k = \text{step}(z_{k-1}, S_t)$). Reasoning occurs in latent state representations without text token emission.

Verified Published Benchmark Evidence

On the public ARC-AGI-1 evaluation set, published research results for BDH-CQ report:

- **Model Scale:** Approximately 150 million parameters (150M).
- **Reasoning Accuracy:** 29.5% pass@2 on the public ARC-AGI-1 evaluation set.
- **Inference Cost:** \$0.0007 per task (under one-tenth of a cent).
- **Citation:** Engdahl et al. (Pathway), *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*, arXiv:2608.09888 (August 2026).

Transparency Demarcation: This result is cited directly from primary published literature as a reported cost-efficiency Pareto point on the public benchmark. It is not presented as the highest raw-accuracy system on ARC-AGI, nor is the educational toy simulator in this repository an implementation of BDH-CQ.

14. Comparing Fixed-State Memory with Attention/KV Caches

To place evolving states in proper context, we compare five distinct sequence-processing paradigms:

Dimension	Standard Transformer (KV Cache)	Recurrent Vector (RNN / GRU)	Linear Attention / SSM (RetNet, Mamba)	Dragon Hatchling (BDH)
Working Memory Mechanism	External buffer of uncompressed key-value vectors	Single hidden vector $h_t \in \mathbb{R}^d$ updated non-linearly	Matrix state $S_t \in \mathbb{R}^{d \times d}$ or selective state parameters	Dynamic synaptic plasticity over scale-free particle graph
Memory Scaling	Linear $O(T \cdot d \cdot L)$ in sequence length T	Fixed $O(d)$ with respect to sequence length T	Fixed $O(d^2)$ with respect to sequence length T	Fixed with respect to sequence length T for fixed config
Inference Step Cost	$O(T)$ Memory-bandwidth bound (scales with context)	$O(d^2)$ Compute bound (constant per step)	$O(d^2)$ Compute bound (constant per step)	Fixed with respect to sequence length T
Information Retention	Exact lossless token storage within context limit	Severe bottleneck (compressed into d scalar values)	Linear superposition; subject to cross-talk interference	Dynamic synaptic adaptation; bounded by latent capacity
Long-Horizon Failure	VRAM out-of-memory (OOM) exhaustion	Vanishing gradients / catastrophic forgetting	Rank saturation & cross-talk noise ($T > d$)	Synaptic saturation in finite latent workspace

Scientific Precision Rule: A fixed-size state keeps the state representation fixed with respect to sequence length for a fixed model configuration, rather than allocating a new memory slot for every token. This does not imply unlimited information capacity or zero information loss.

15. What This Toy Model Does NOT Claim

Intellectual honesty requires establishing what our educational simulator does not do:

- 1. Not an Official BDH Implementation:** The simulator is an educational matrix model designed to make the linear algebra of recurrence visible. It does not execute the multi-layer, scale-free particle graph algorithm of BDH.
- 2. Not a Replacement for Softmax Attention:** The educational model demonstrates that unconstrained linear associative recall cannot achieve the sharp, non-linear selective focus of Softmax attention when key distributions are non-orthogonal.
- 3. No Infinite Context Without Loss:** The model proves that while fixed-size states can ingest arbitrarily long sequences without memory allocation growth, information retrieval degrades along a predictable capacity curve once sequence length exceeds available subspace dimensions.
- 4. No Universal OOM Immunity:** While a fixed-size state avoids sequence-length OOM from cached token vectors, an overall system can still exhaust memory due to activation storage, batch size, or parameter footprint.

16. Evidence and Research Context

The concepts taught in this project are grounded in peer-reviewed and preprint literature published between 1982 and 2026:

- **Linear Attention as Fast Weights:** Schlag, Irie, & Schmidhuber (ICML 2021) mathematically established that unnormalized linear Transformers are equivalent to fast-weight recurrent associative memories updating matrix states via outer products.
- **Retention Networks:** Sun et al. (Microsoft Research, 2023) demonstrated that multi-scale retention with explicit exponential decay ($\lambda < 1.0$) achieves constant inference memory and linear-time training.
- **Selective State Spaces:** Gu & Dao (2023, Mamba) showed that making recurrent transition matrices input-dependent allows models to selectively filter relevant tokens into compressed states.
- **Synaptic Plasticity & Recurrent Reasoning:** Kosowski et al. (2025, arXiv:2509.26507) and Engdahl et al. (2026, arXiv:2608.09888) demonstrated that Hebbian attention ($Q=K$) and latent recurrent reasoning achieve remarkable cost-efficiency Pareto trade-offs on challenging reasoning benchmarks.

17. Reproducibility

Every interactive figure, metric, and table in this project is 100% reproducible:

- **Deterministic PRNG:** All synthetic tasks and vector generations use the Mulberry32 pseudorandom generator with fixed seeds (seed 42 for standard benchmarks, seed 555 for diagnostic questions).
- **Automated Unit Testing:** 12 automated unit tests in Vitest verify vector normalization, outer-product algebra, Frobenius norm conservation, SNR decomposition, and singular spectrum computation.
- **Open-Source Codebase:** Complete source code is available on GitHub under the MIT License, with step-by-step reproduction instructions in `README.md` and detailed experimental logs in `EXPERIMENTS.md`.

18. AI Assistance and Disclosure

In full accordance with competition transparency guidelines, AI tools were utilized during development for:

- Scaffolding initial React 19 + TypeScript + Tailwind CSS component templates.
- Assisting with vector linear algebra helper routines and automated unit test assertions.
- Editorial feedback on technical writing clarity and formatting.

Human / Team Ownership: The human team directed the conceptual scope, selected the central falsifiable claim, structured the 10-module pedagogical progression, audited every mathematical formula and citation against primary sources, established the university laboratory visual design, and executed all verification builds and tests.

19. References

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